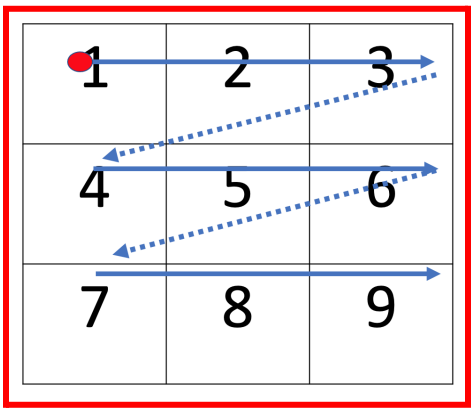
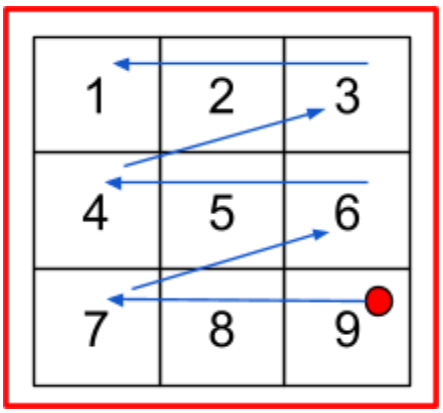
Programming Fundamentals
Lab-12

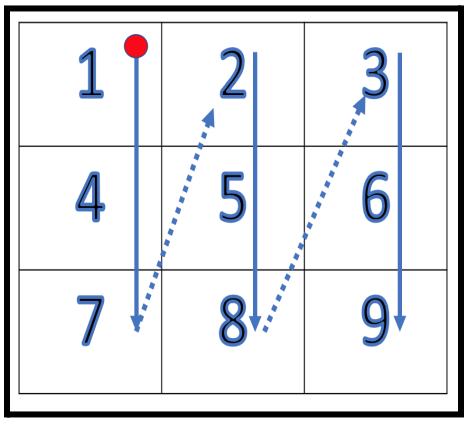
Some Practice with 2D Matrix Traversals

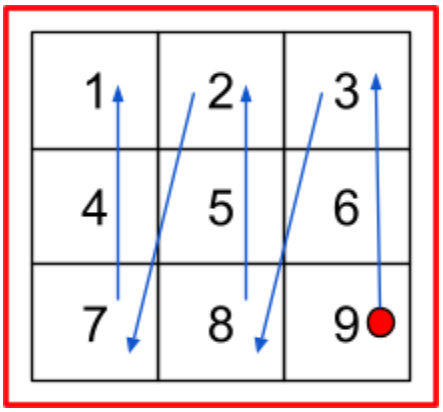
Task: Traversals and mappings from R^2 to R dimensional world

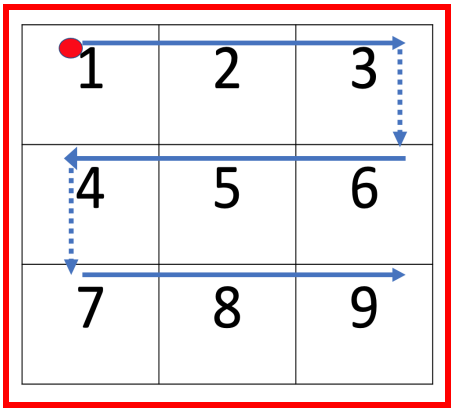
Given a **n x n** matrix **M**, return an array of all the elements of the matrix in the arrow orders. Make 10 separate functions, which convert a two dimensional matrix into single dimensional array in the following order

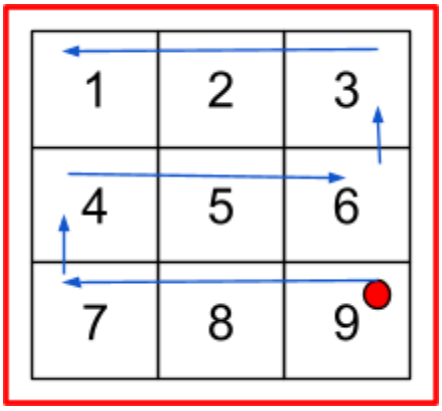


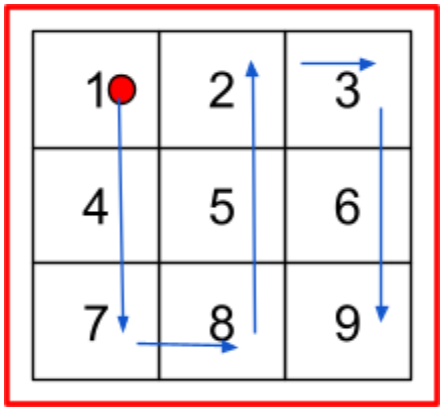


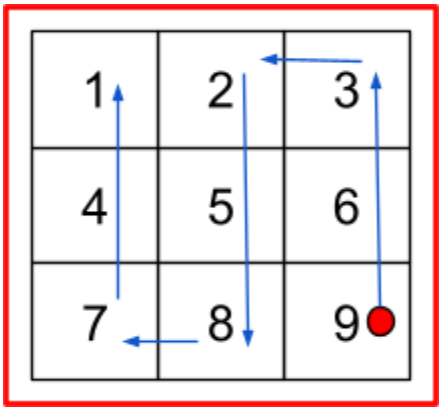


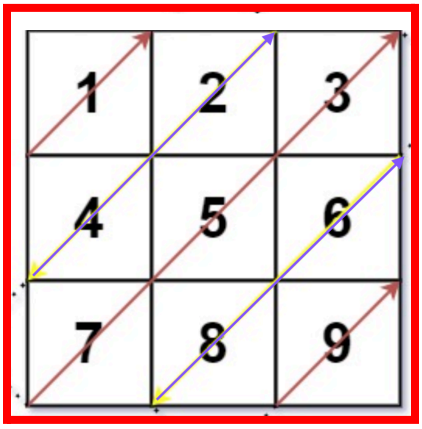


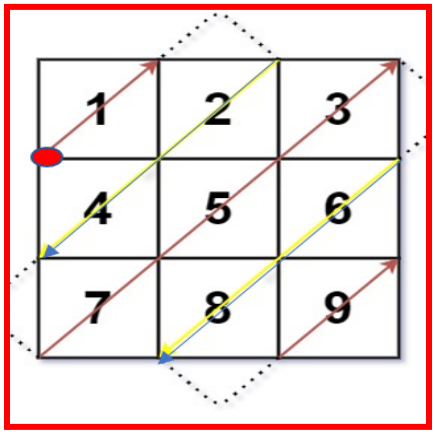




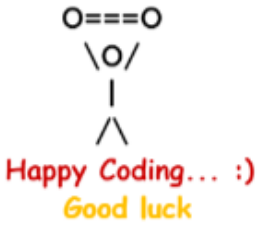








Application: Matrix Column and Row Sum

Write a function (or two separate function) that takes a 2D matrix M and returns two 1D array (or returns one dimensional arrays separately) that has sum of each column on it's indices.

Sample Input	Sample Output
M: 1 1 1 1 1 2 2 2 2 2 3 3 3 3 3 4 4 4 4 4 1 2 3 4 5	Column Sums list: 11 12 13 14 15 Row Sums list: 5 10 15 20 15